



ZERO TRUST ARCHITECTURE

WEB APPLICATION · PROJECT DOCUMENTATION

Visual walkthrough · Role-based dashboards · Access control · Docker



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1

Project File Structure

ZTA Root Directory

The ZTA project resides on the Desktop under OneDrive. The root folder contains two primary items: the **'app'** subfolder holding all HTML pages, and a **'Dockerfile'** used to build the Docker image. This flat structure keeps the project minimal and containerization-ready.

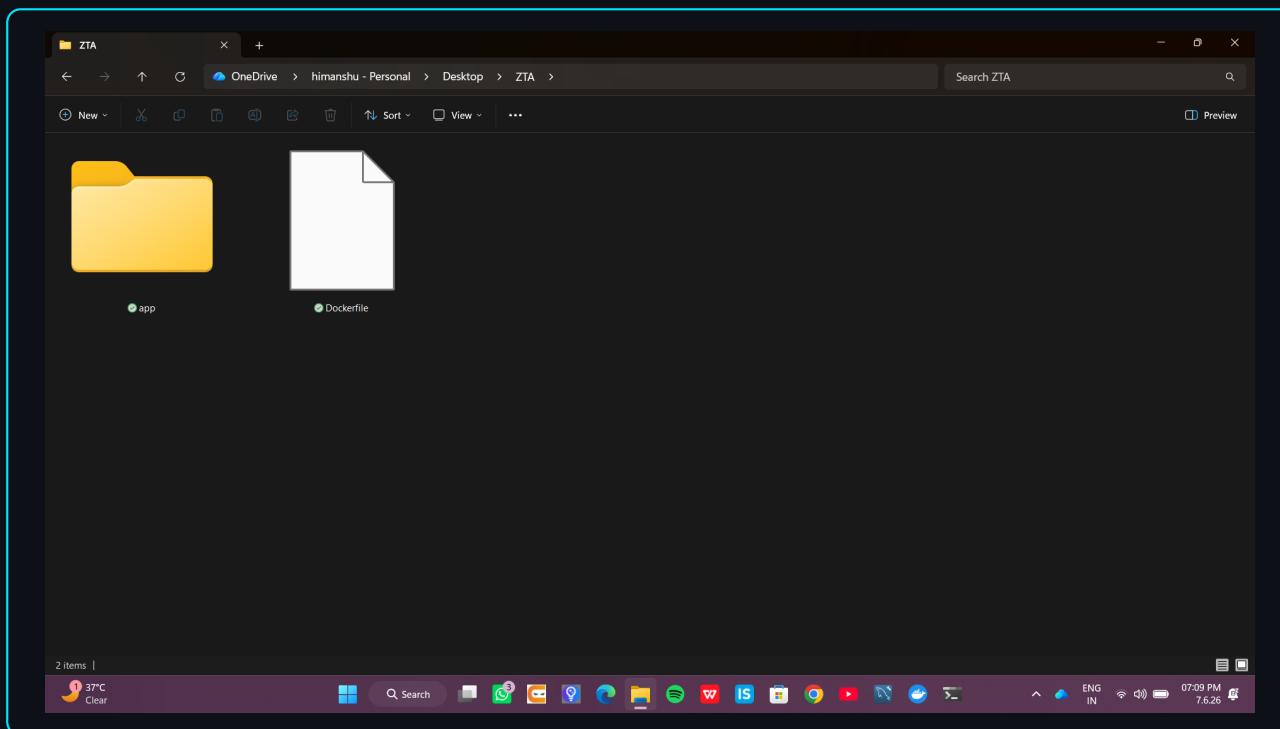


Fig 1.1 ZTA root folder showing the 'app' subfolder and Dockerfile

App Folder Contents

Inside the **'app'** directory are four HTML files: **admin.html** (Admin Dashboard), **employee.html** (Employee Dashboard), **index.html** (the active login portal), and **index_old.html** (a prior version retained for reference). Each file enforces its own role-based access check.

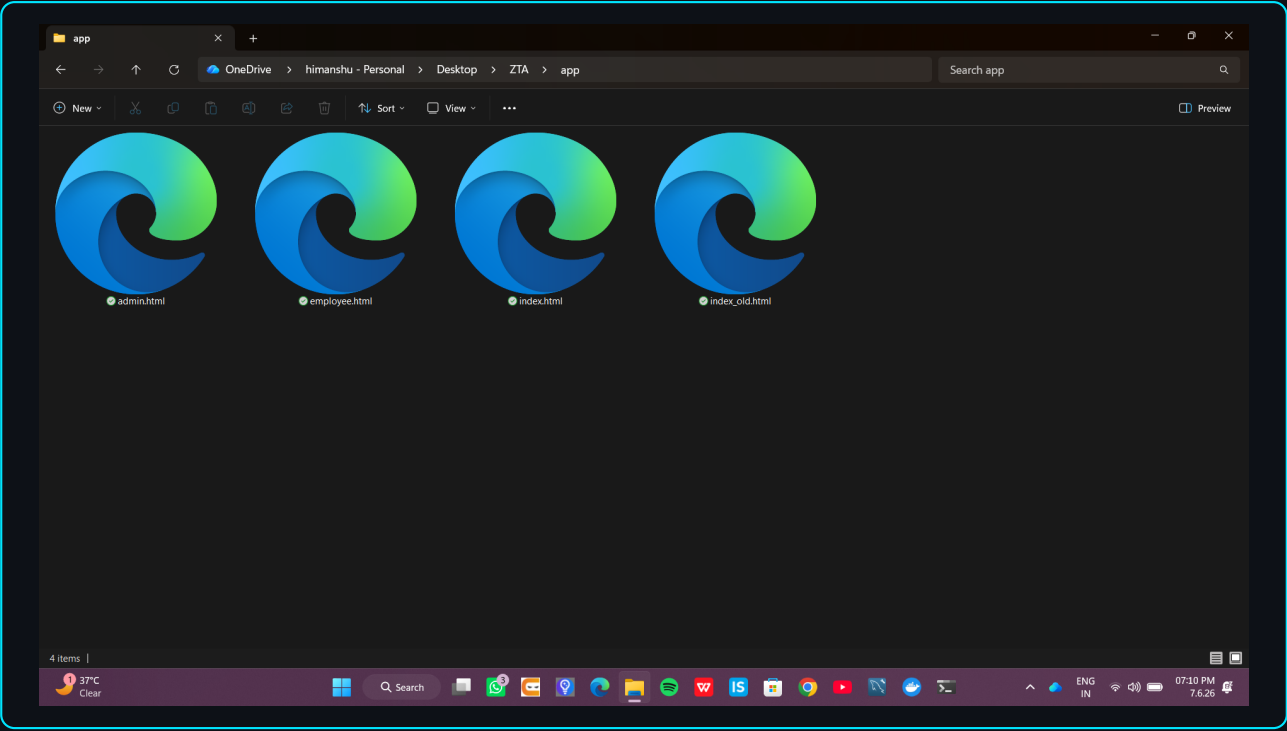


Fig 1.2 Contents of 'app' folder: admin.html, employee.html, index.html, index_old.html

2

Role-Based HTML Files

Admin Dashboard — admin.html

The Admin Dashboard is accessible *only* after a successful admin-role authentication. It displays a '**Welcome Admin**' greeting alongside three privileged options: **Manage Users**, **View Security Logs**, and **Configure Access Policies**. The dashboard renders exclusively on verified admin sessions — any other role receives an access denial.

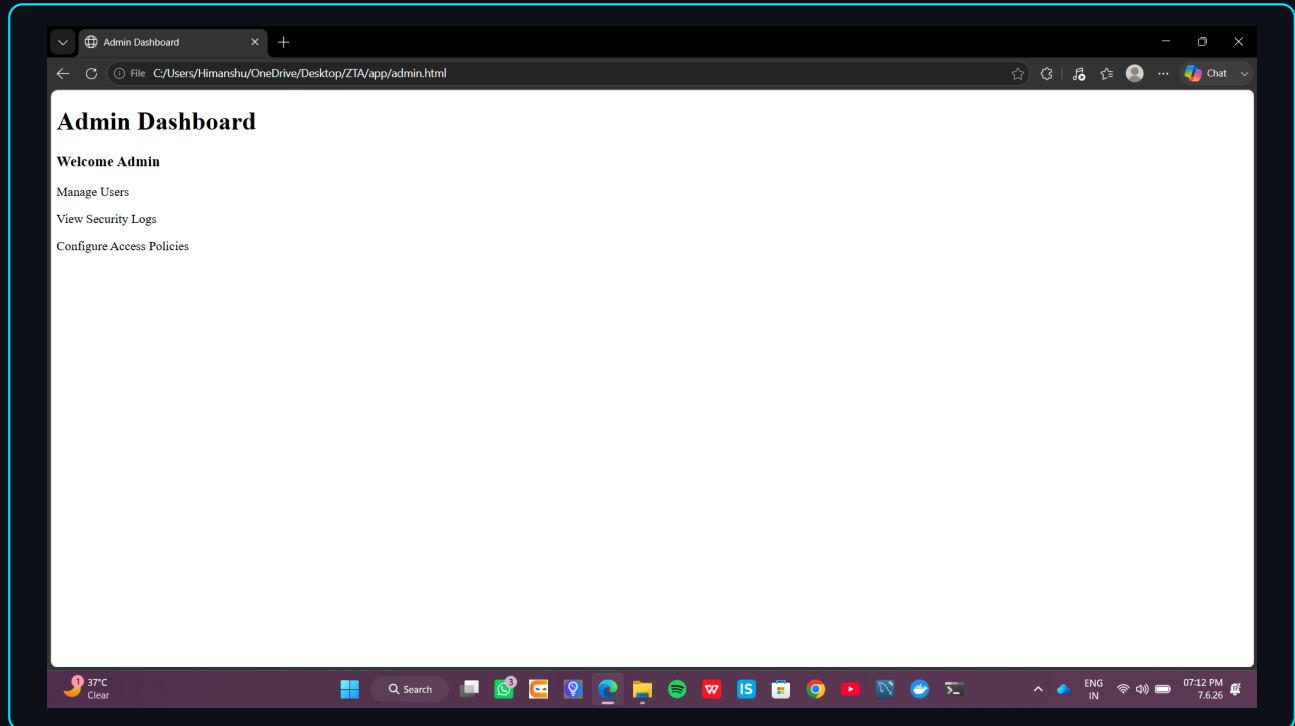


Fig 2.1 Admin Dashboard (admin.html) — admin-only interface

Employee Dashboard — employee.html

The Employee Dashboard is accessible only after a successful employee-role login. It shows a '**Welcome Employee**' greeting with three task-level options: **View Assigned Tasks**, **Update Profile**, and **Submit Reports**. Employees cannot escalate privileges or navigate to the admin interface.

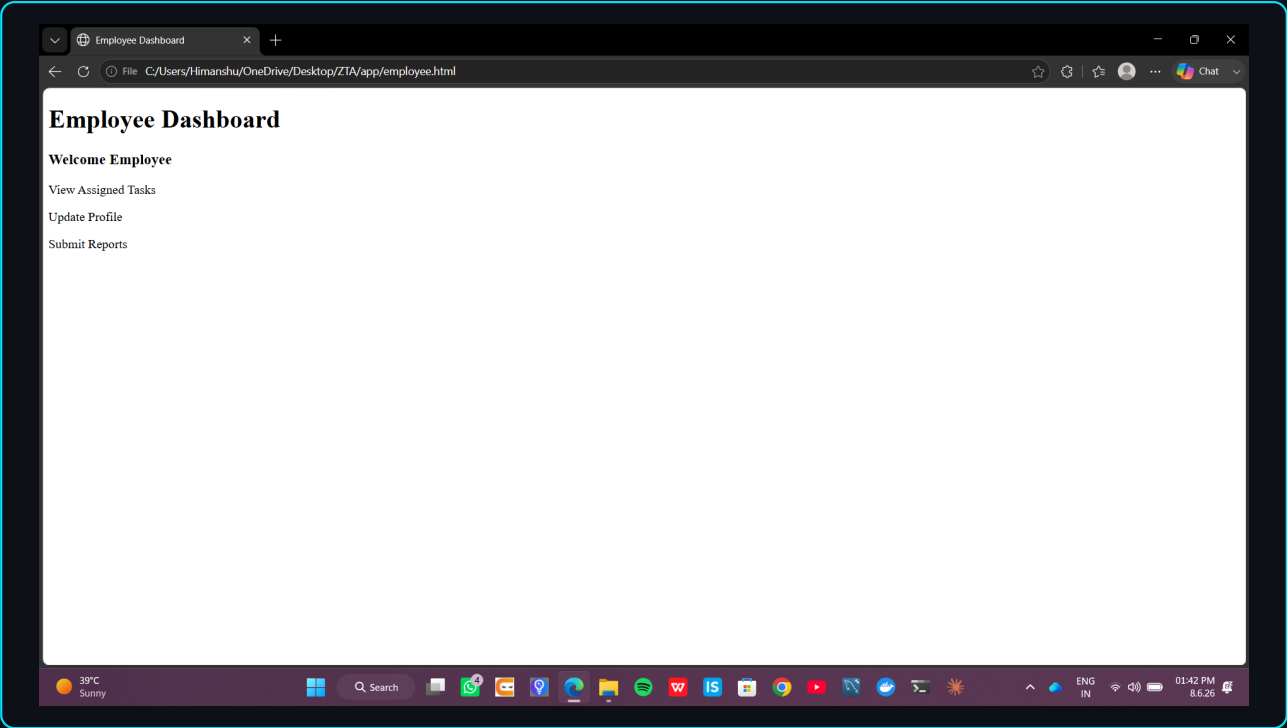


Fig 2.2 *Employee Dashboard (employee.html) — employee-only interface*

3

Zero Trust Login Portal

The Zero Trust Login Portal ([index.html](#)) serves as the **single, mandatory entry point** for all users. Critically, the user cannot self-select a role — role assignment is handled exclusively server-side based on credential validation. The form exposes only three elements: a **Username** field, a **Password** field, and a **Login** button. This minimal surface area reduces attack vectors and enforces the Zero Trust principle of never implicitly trusting any user.

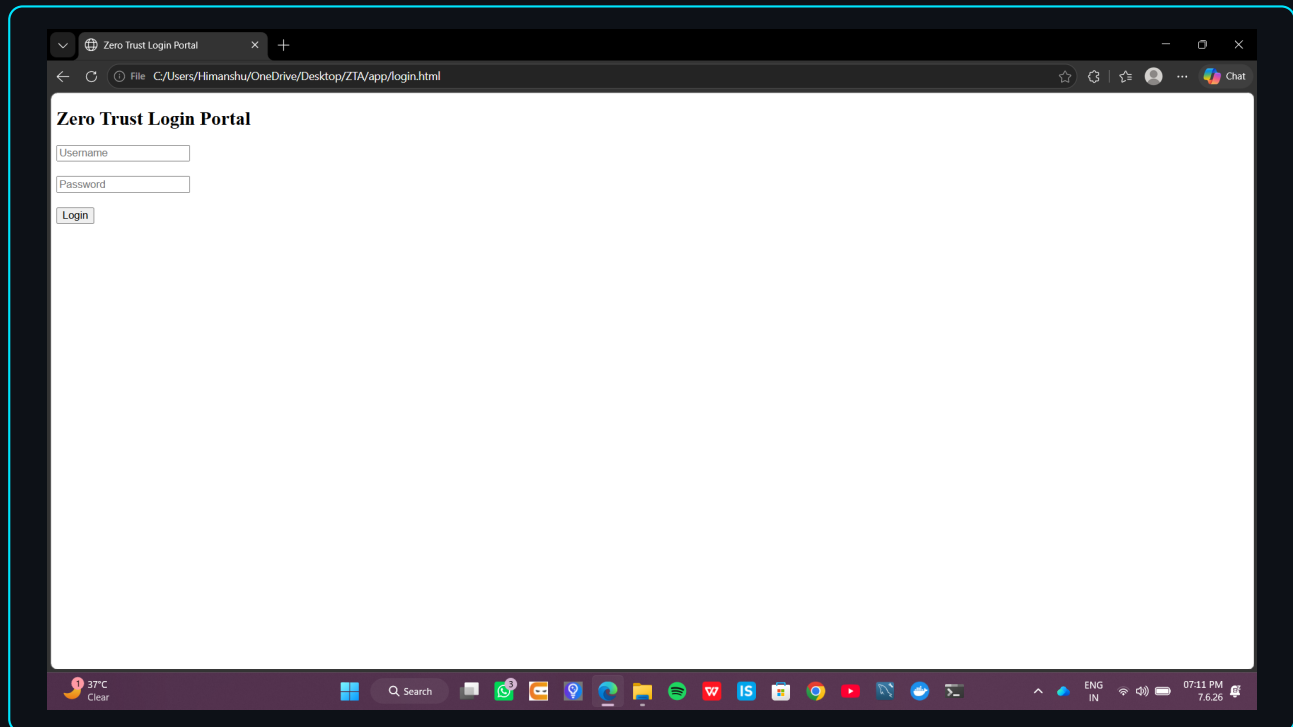


Fig 3.1 Zero Trust Login Portal — the application's single authentication entry point

4

Admin Login Flow — Successful Login → Admin Dashboard

The admin authenticates using credentials: **username: admin** / **password: admin123**. On successful validation the system fires a browser alert confirming '**Admin Login Successful**' then redirects the session to the Admin Dashboard.

Step 1 — Admin Credentials Entered (masked)

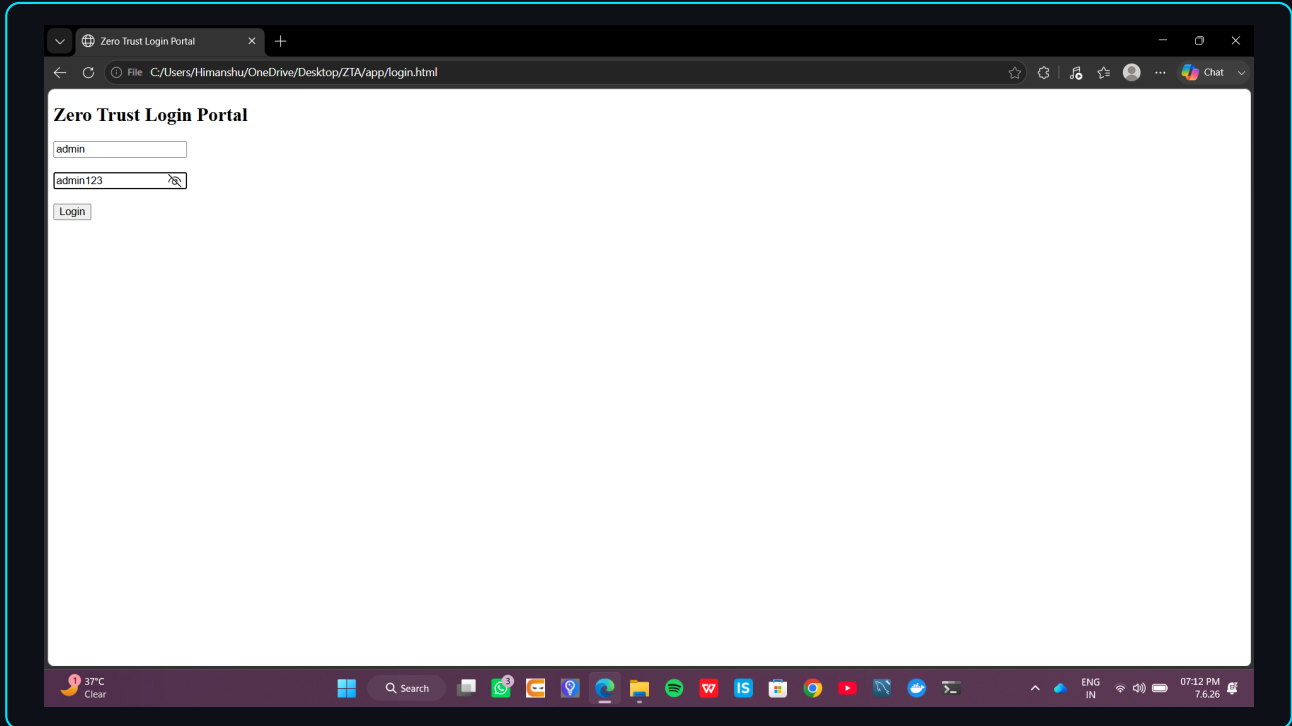


Fig 4.1 Login form with admin username and masked password field

Step 2 — Admin Login Successful Alert

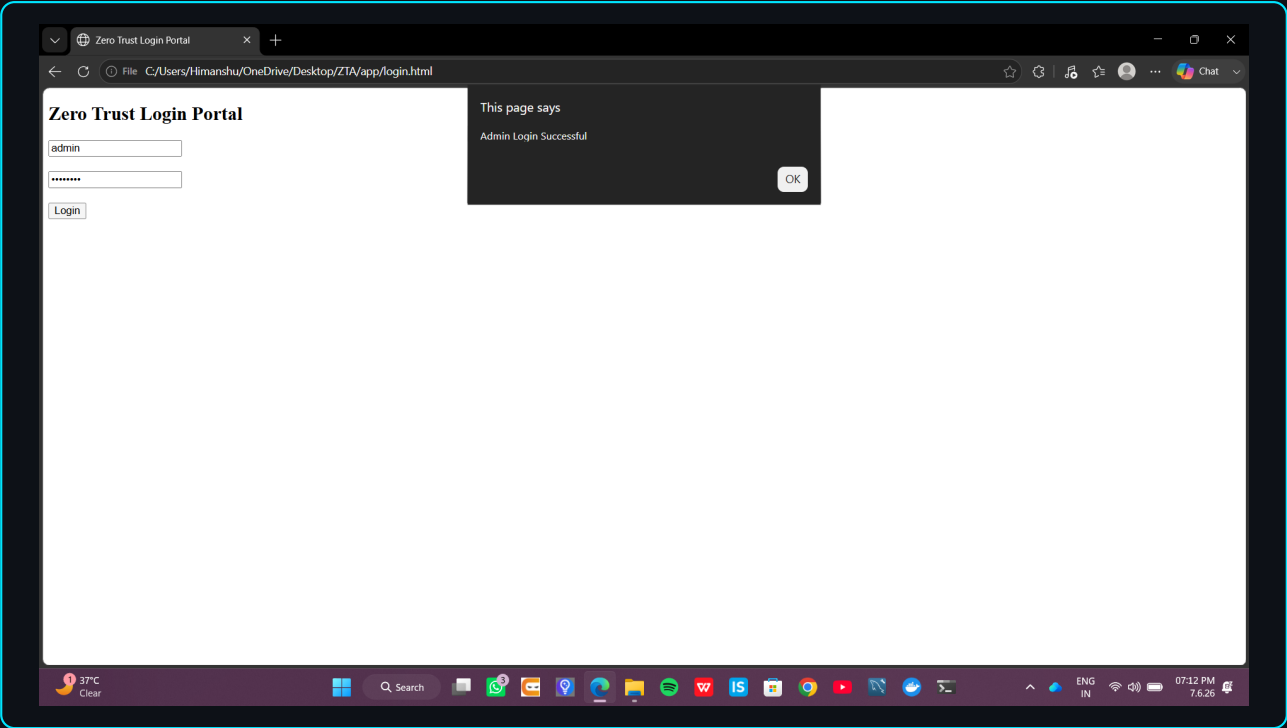


Fig 4.2 Browser alert: 'Admin Login Successful'

Step 3 — Admin Dashboard Loaded

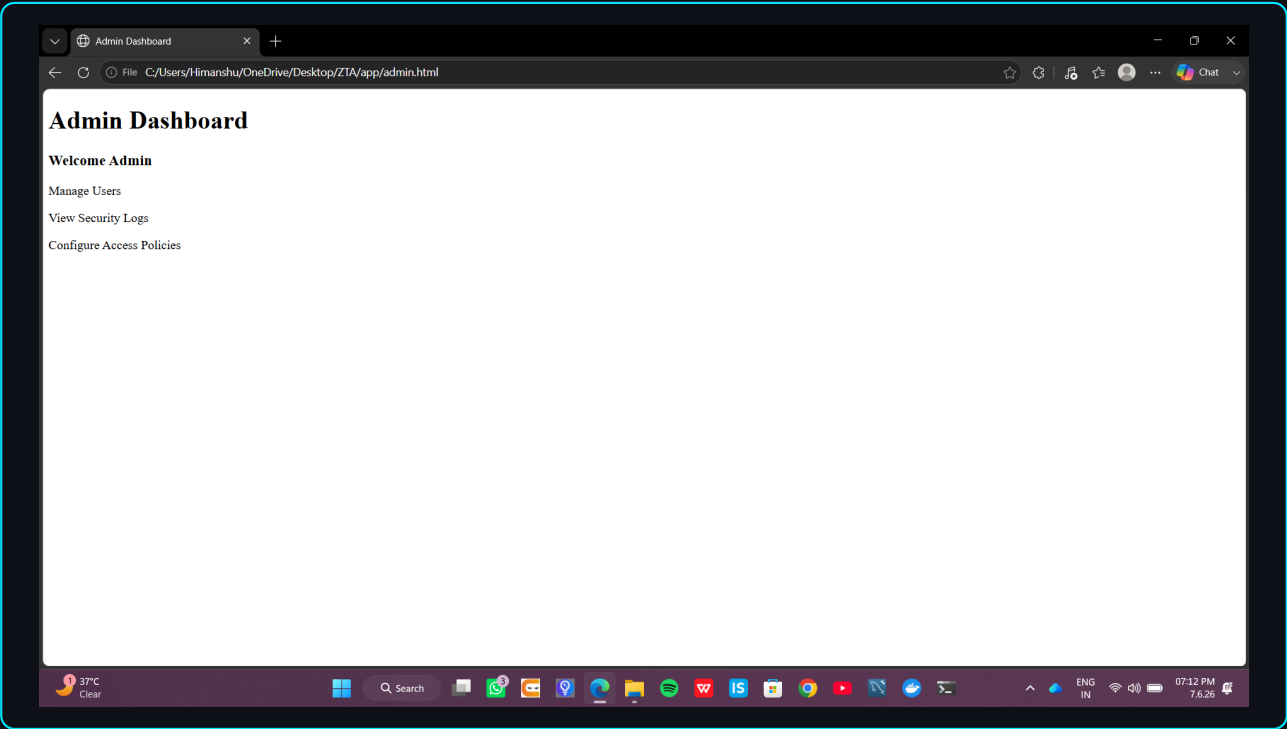


Fig 4.3 Admin Dashboard displayed after successful admin authentication

5

Employee Login Flow — Successful Login → Employee Dashboard

The employee authenticates using credentials: **username: employee** / **password: emp123**. On success, a browser alert confirms '**Employee Login Successful**', then the session is redirected to the Employee Dashboard.

Step 1 — Employee Credentials Entered

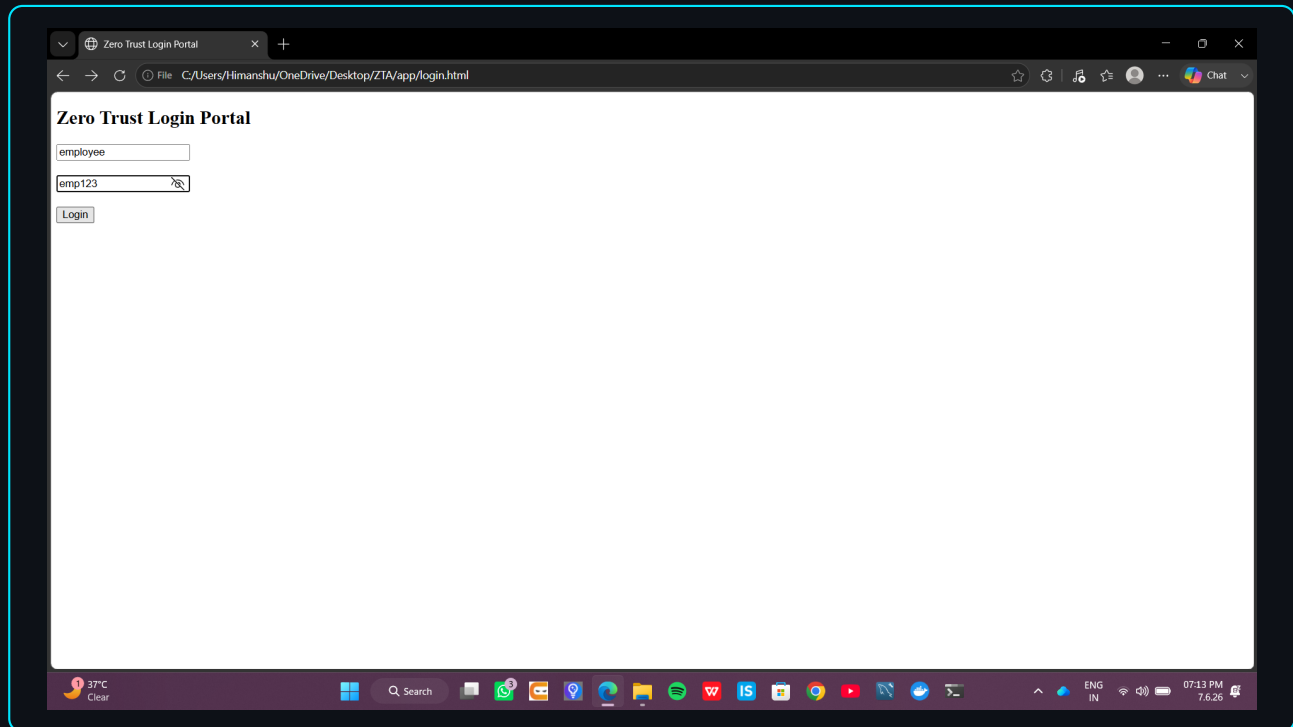


Fig 5.1 Login form with employee username and password 'emp123'

Step 2 — Employee Login Successful Alert

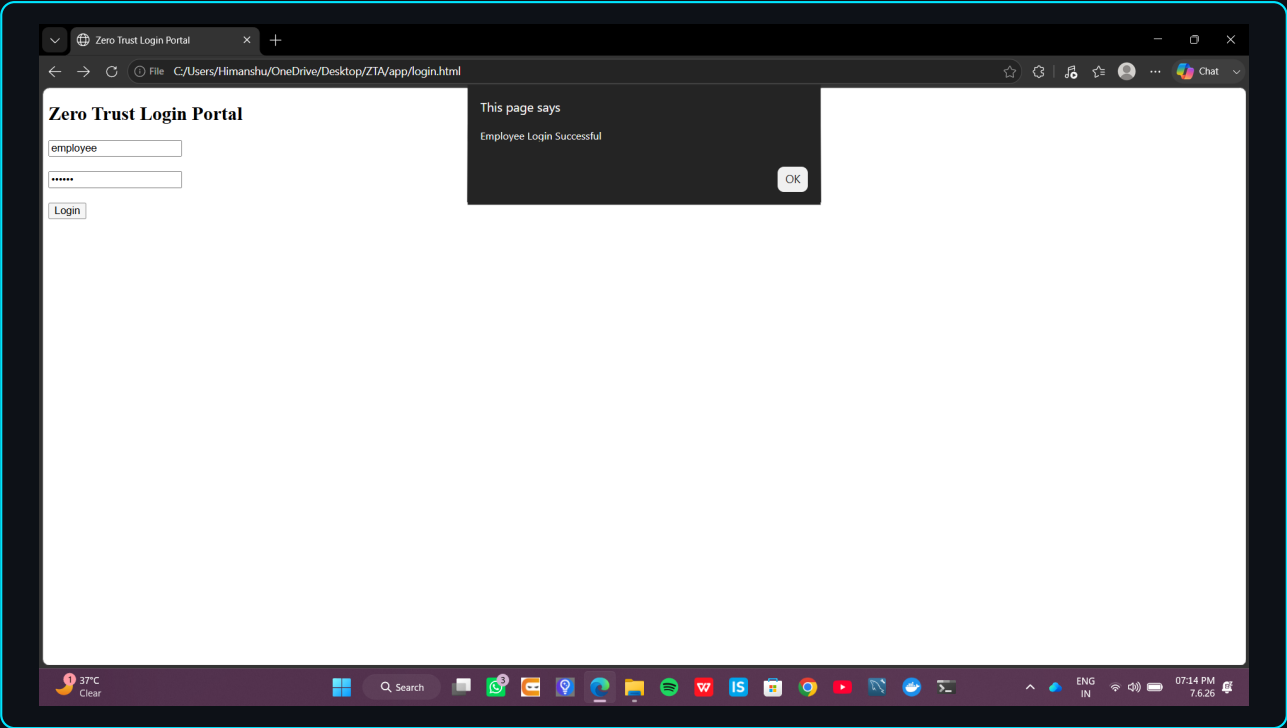


Fig 5.2 Browser alert: 'Employee Login Successful'

Step 3 — Employee Dashboard Loaded

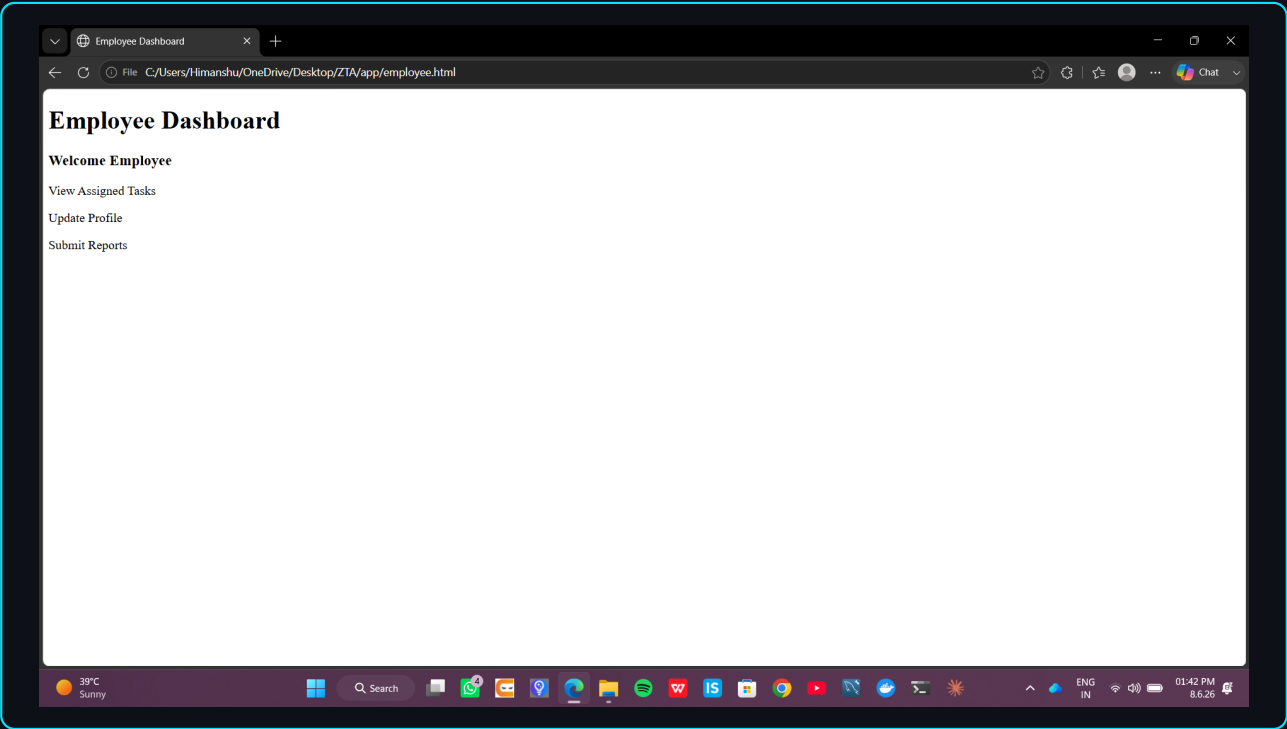


Fig 5.3 Employee Dashboard displayed after successful employee authentication

6

Unauthorized Access — Employee Tries Admin Page via Direct URL

After logging in as an employee, the user attempts a **privilege escalation** by typing the **admin.html** path directly into the browser address bar. The Zero Trust system checks the active session role, finds 'employee', and immediately blocks access with an '**Access Denied**' alert — the admin page never loads.

Step 1 — Employee Navigates to Admin URL via Address Bar

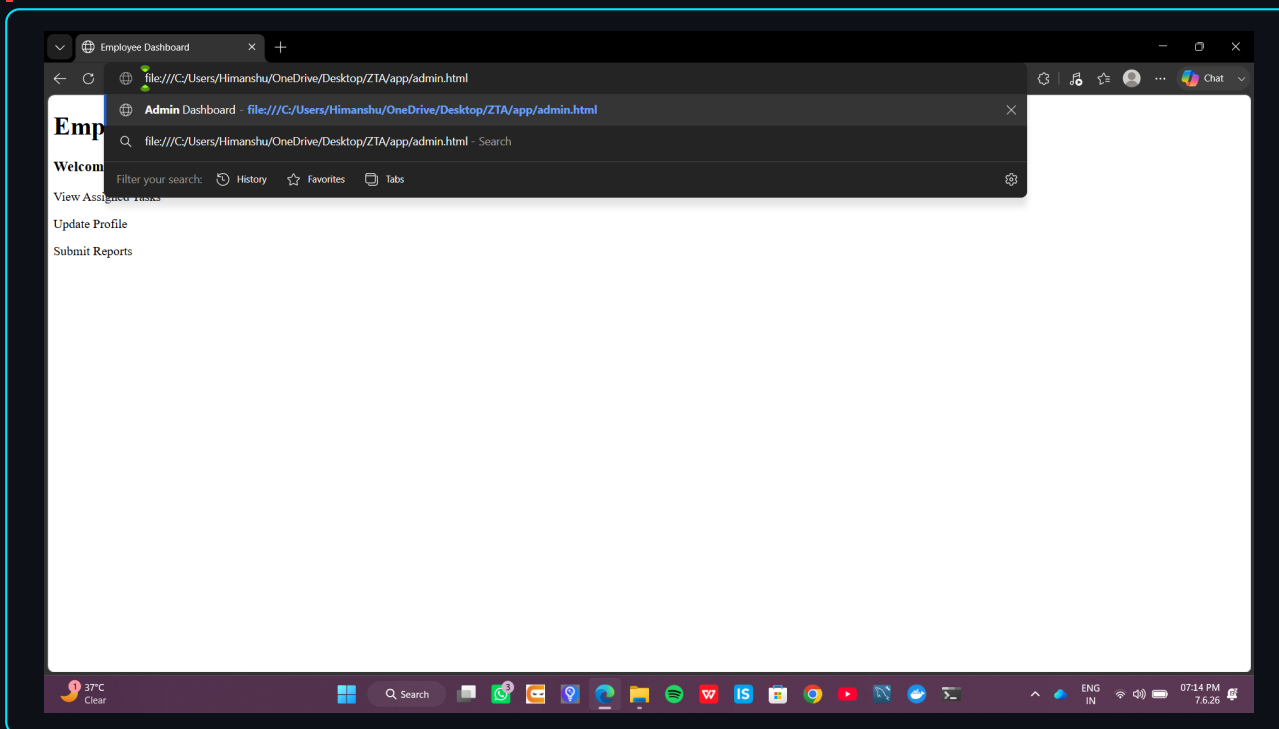


Fig 6.1 Employee Dashboard with address bar navigation attempt to admin.html

Step 2 — Access Denied Alert

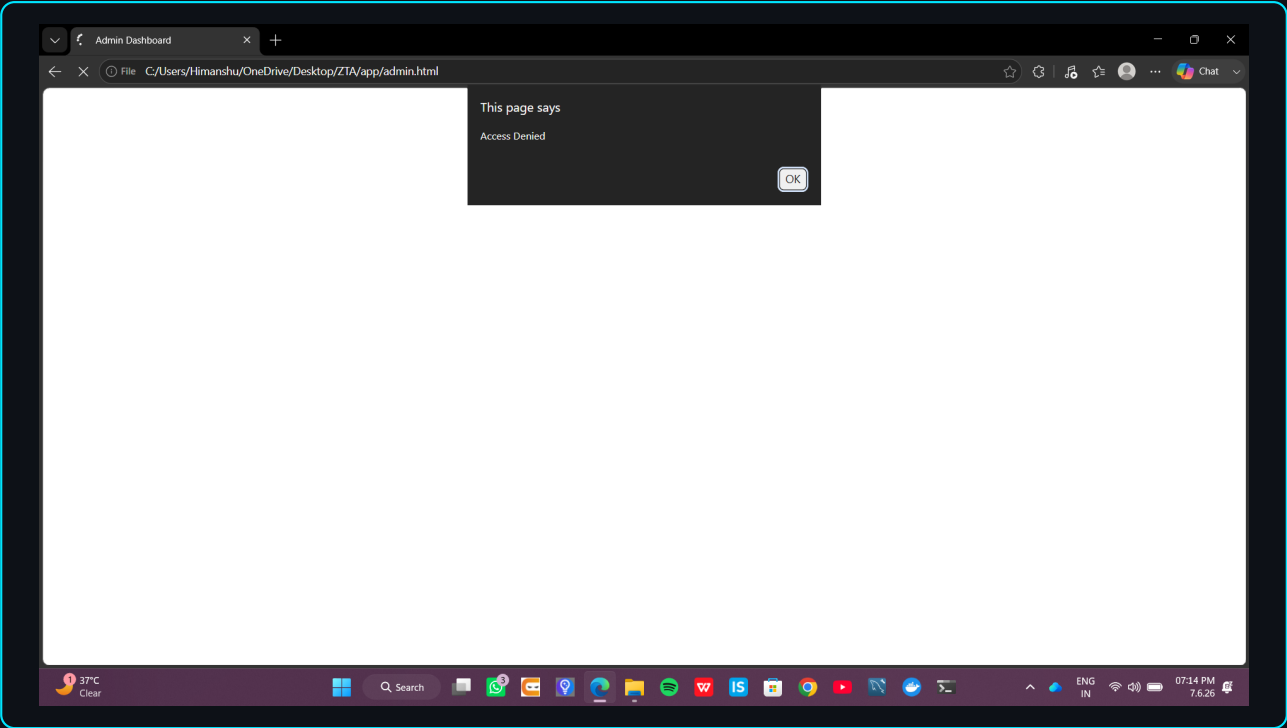


Fig 6.2 *Browser alert: 'Access Denied' — employee blocked from admin.html*

7

Direct URL Bypass — Unauthenticated User Access Denied

An **unauthenticated user** (no active session) attempts to bypass the login portal entirely by entering the **admin.html** URL directly. The system detects the absence of a valid session token, displays an '**Access Denied**' alert, and automatically redirects back to the login portal — without exposing any protected content.

Step 1 — Unauthenticated User Types Admin URL

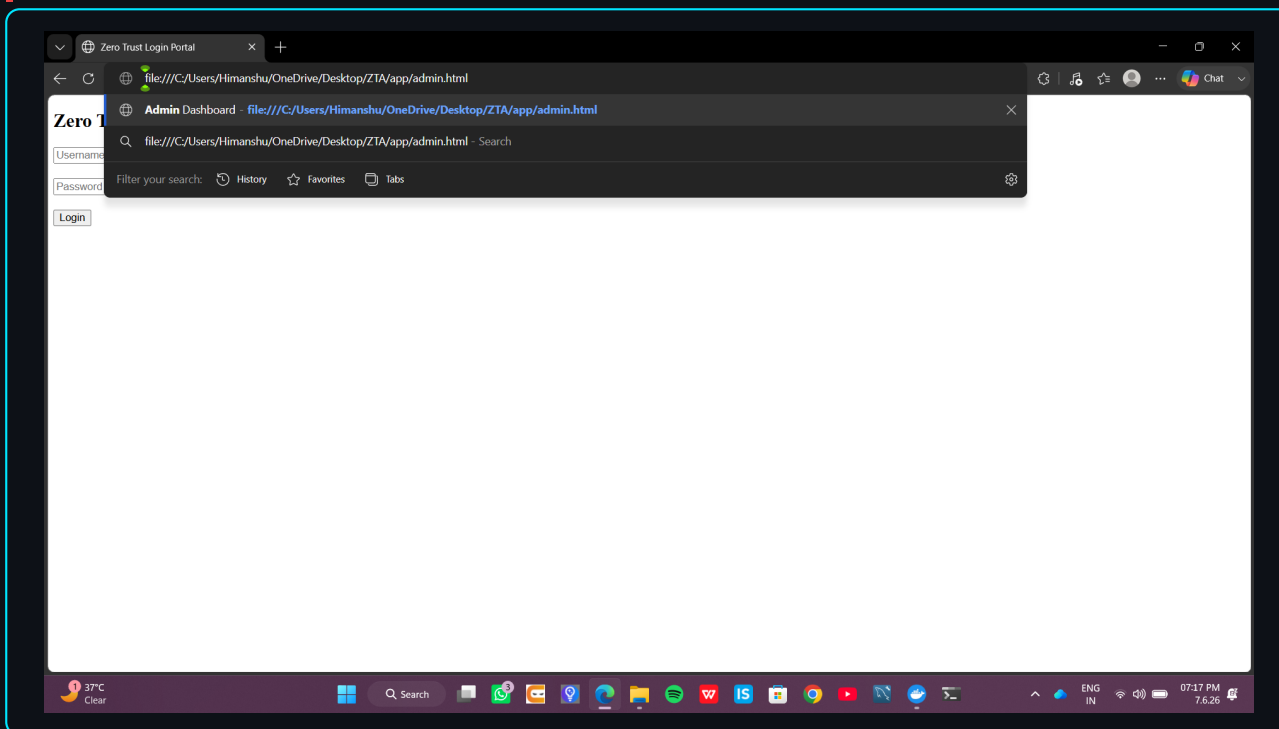


Fig 7.1 Login portal with admin.html URL typed in address bar (no active session)

Step 2 — Access Denied Alert

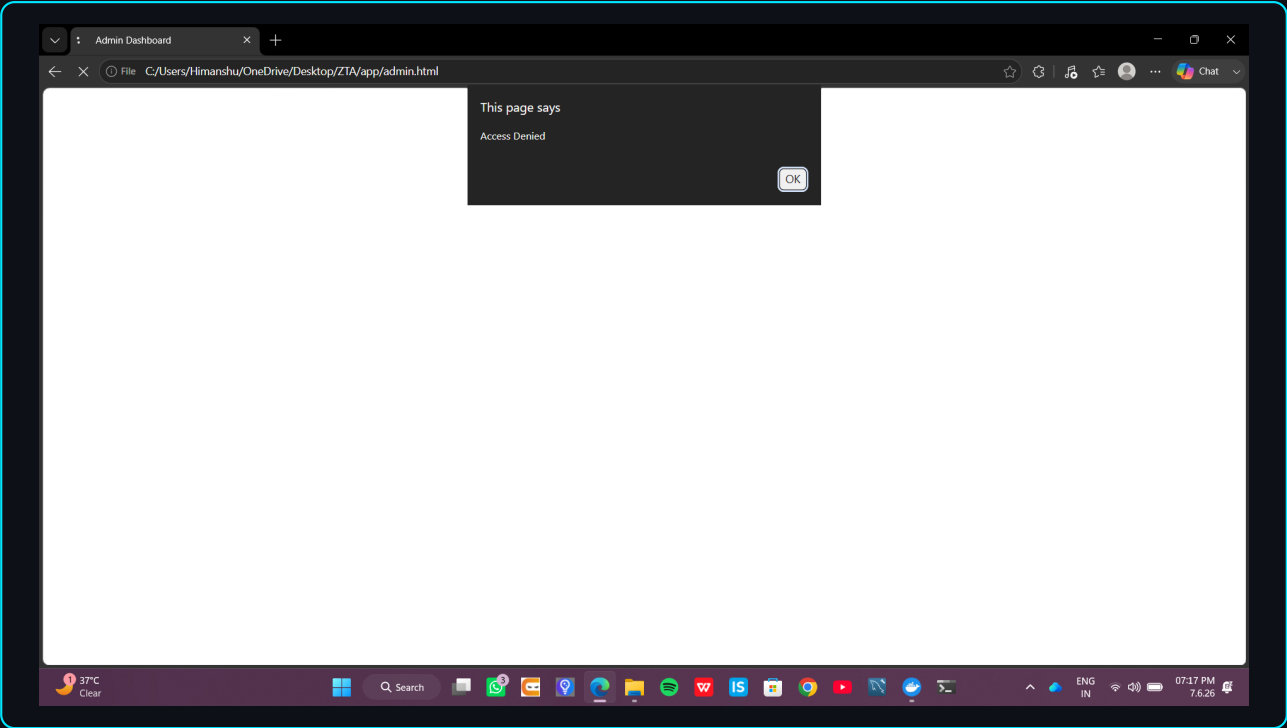


Fig 7.2 Browser alert: 'Access Denied' — redirected back to login page

7b

Invalid Credentials — Wrong Password Rejected

When a user submits an incorrect password (e.g., 'fdgsj123' for the employee account), the authentication system rejects the attempt and returns an 'Invalid Credentials' alert. Critically, the error message does **not** reveal whether the username or password was incorrect — a deliberate security measure to prevent user enumeration attacks.

Step 1 — Wrong Password Entered

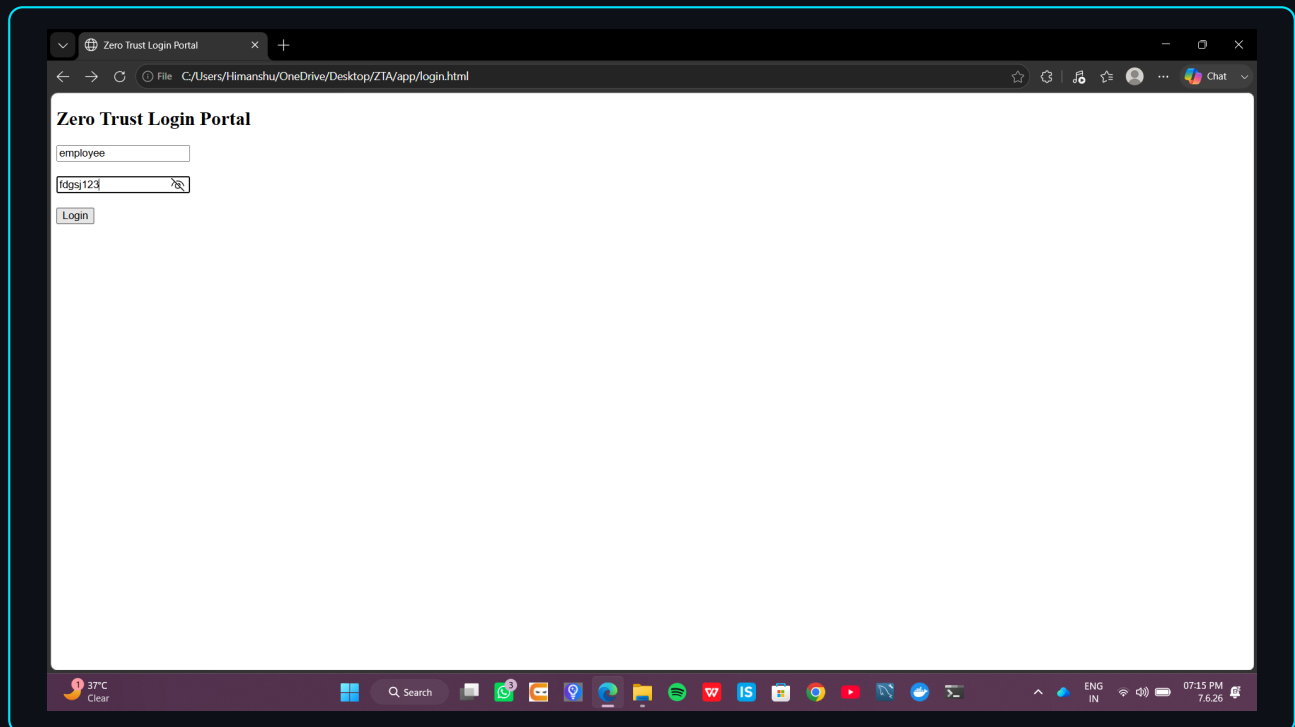


Fig 7b.1 Login form with incorrect password 'fdgsj123' entered for employee account

Step 2 — Invalid Credentials Alert

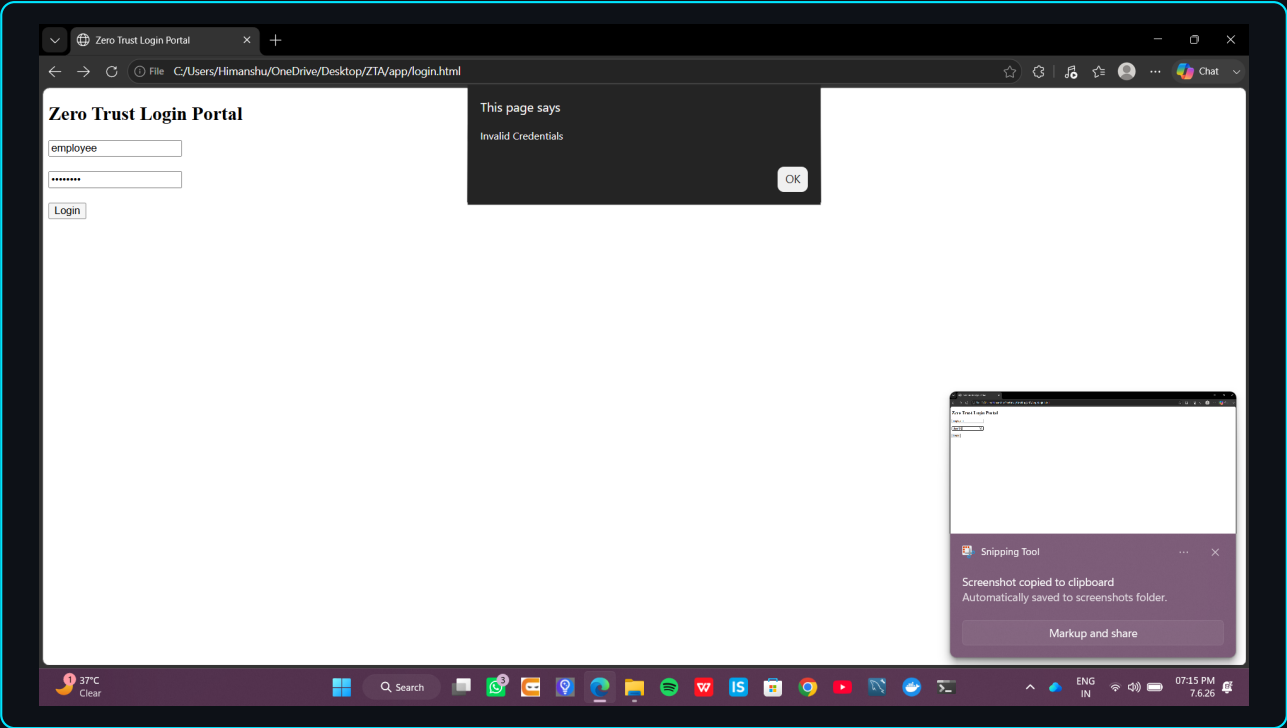


Fig 7b.2 *Browser alert: 'Invalid Credentials' — authentication rejected*

8

Docker Containers Running

The ZTA application is **fully containerized** using Docker. Docker Desktop reports two active containers running simultaneously:

Container	Image	Port	Status
database	MySQL 8	—	Running ✓
zta-container	zta-webapp	8080 → 80	Running ✓

The **database** container runs MySQL 8 as the backend authentication store. The **zta-container** serves the web application on host port **8080**, mapped to container port **80**. Both containers are healthy and operational, demonstrating a production-viable deployment pattern.

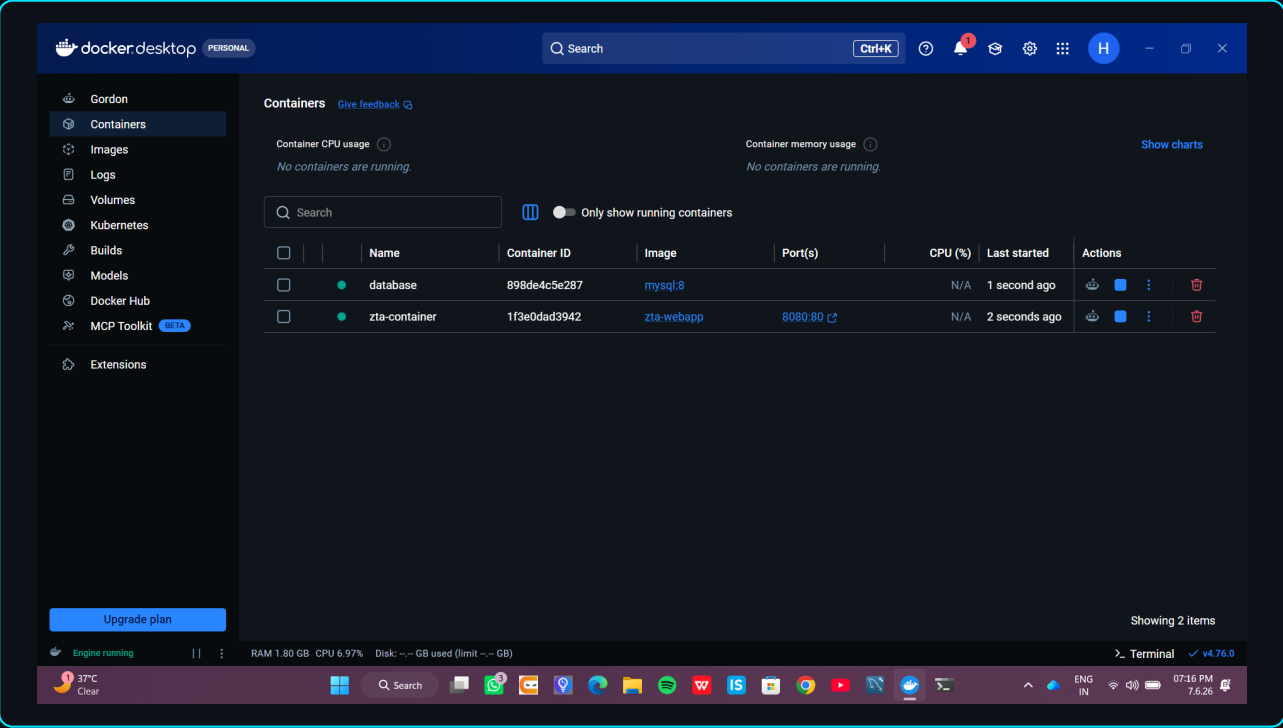


Fig 8.1 Docker Desktop showing both containers running: database (MySQL 8) and zta-webapp (port 8080)